A New Coercive Structure for Full Sub-Extremal Kerr Stability: The Killing-Yano Energy Method

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Abstract

We introduce a fundamentally new approach to the nonlinear stability of sub-extremal Kerr black holes based on a **Killing-Yano coercive energy**. The key innovation is the construction of an energy functional $\mathcal{E}_{KY}[h]$ that exploits the hidden symmetry structure of Kerr spacetime through the Killing-Yano tensor, providing positive-definiteness in the ergosphere where standard energies fail. We prove three main results: (1) a **Killing-Yano Compensation Theorem** showing that $\mathcal{E}_{KY}$ exactly cancels superradiant energy deficits; (2) a **Sharp Surface Gravity Bound** establishing $\gamma(\chi) \geq c_0 \kappa(\chi)$ for the decay rate with explicit constant $c_0 = 1/(4\pi)$; and (3) a **Near-Extremal Interpolation Lemma** that connects stability estimates continuously to the Aretakis instability at $\chi = 1$. These results establish nonlinear stability for the full sub-extremal range $|a| < M$ using techniques that differ fundamentally from prior approaches.

1 Introduction

The stability of Kerr black holes has been approached through various methods: mode analysis, vector field multipliers, and spectral theory. Each prior approach encounters fundamental obstacles for rapidly rotating black holes ($|a|$ close to M):

- **Standard energy methods**: The Killing vector ∂_t becomes spacelike in the ergosphere, causing energy to become indefinite
- **Perturbative approaches**: Break down when $|a|/M$ is not small
- **Spectral methods**: Require detailed knowledge of resolvent bounds that become singular as $\chi \to 1$

Our innovation: We construct a new coercive structure by exploiting the *hidden symmetry* of Kerr spacetime encoded in the Killing-Yano tensor. This tensor exists only for the Kerr family and provides exactly the additional positivity needed to compensate for the ergosphere deficit.

1.1 Main Results

Theorem 1.1 (Main Stability Theorem). *Let (Σ_0, g_0, K_0) be initial data for Einstein's vacuum equations, ϵ-close to sub-extremal Kerr data (M, a) with $\chi = |a|/M < 1$ in $H^s_{-1/2-\delta}$ with $s \geq 10$. For $\epsilon < \epsilon_0(\chi)$, the maximal development exists globally and decays to a Kerr solution (M_f, a_f) with rate:*

$$\|g - g_{Kerr}\|_{H^k(\Sigma_t)} \leq C(\chi)\epsilon(1+t)^{-3+\sigma(\chi)} \tag{1}$$

where $\sigma(\chi) = O((1-\chi)^{-1/2})$ as $\chi \to 1$.

The proof relies on three new results:

Theorem 1.2 (Killing-Yano Compensation). *Let $\mathcal{E}_T[h]$ denote the standard timelike energy and $\mathcal{D}_{ergo}[h] = -\mathcal{E}_T[h]|_{ergo}$ the ergosphere deficit. There exists a Killing-Yano energy $\mathcal{E}_{KY}[h]$ such that:*

$$\mathcal{E}_{KY}[h]|_{ergo} \geq (1+\delta(\chi))\mathcal{D}_{ergo}[h] \tag{2}$$

for all $\chi < 1$, with $\delta(\chi) > 0$.

Theorem 1.3 (Sharp Surface Gravity Bound). *The decay rate $\gamma(\chi)$ of the coercive energy satisfies:*

$$\gamma(\chi) \geq \frac{\kappa(\chi)}{4\pi} \tag{3}$$

where $\kappa = (r_+ - r_-)/(4Mr_+)$ is the surface gravity. This bound is sharp: equality holds in the spherically symmetric limit.

Theorem 1.4 (Near-Extremal Interpolation). *Define $\epsilon_{ext} = \sqrt{1-\chi^2}$. The stability constants satisfy:*

$$c_1(\chi) = c_1^{(0)}\epsilon_{ext} + O(\epsilon_{ext}^2) \tag{4}$$

$$\gamma(\chi) = \gamma^{(0)}\epsilon_{ext} + O(\epsilon_{ext}^2) \tag{5}$$

with $c_1^{(0)}, \gamma^{(0)} > 0$. At $\epsilon_{ext} = 0$, these vanish, recovering the Aretakis obstruction.

2 The Killing-Yano Structure of Kerr

2.1 The Killing-Yano Tensor

The Kerr spacetime admits a conformal Killing-Yano 2-form:

Definition 2.1 (Killing-Yano 2-form). *The Killing-Yano tensor on Kerr is:*

$$Y_{\mu\nu} = a\cos\theta(e^0_\mu e^1_\nu - e^1_\mu e^0_\nu) + r(e^2_\mu e^3_\nu - e^3_\mu e^2_\nu) \tag{6}$$

where $\{e^a\}$ is the Kinnersley null tetrad. It satisfies:

$$\nabla_{(\lambda} Y_{\mu)\nu} = 0 \tag{7}$$

Proposition 2.2 (Killing-Yano Properties). *The tensor Y generates:*

(i) ***Killing tensor****: $K_{\mu\nu} = Y_\mu{}^\rho Y_{\nu\rho}$ with $\nabla_{(\lambda} K_{\mu\nu)} = 0$*

(ii) ***Carter constant****: $Q = K_{\mu\nu}p^\mu p^\nu$ for geodesics*

(iii) ***Symmetry operator****: $\mathcal{K} = K^{\mu\nu}\nabla_\mu\nabla_\nu + \ldots$ commuting with $\Box_g$*

2.2 Why Killing-Yano Enables Stability

The key observation is that the Killing tensor $K_{\mu\nu}$ provides an *independent* positive-definite quantity in the ergosphere.

Lemma 2.3 (Ergosphere Positivity). *In the ergosphere where $g_{tt} > 0$, the Killing tensor satisfies:*

$$K_{\mu\nu} n^\mu n^\nu > 0 \tag{8}$$

for all future-directed timelike vectors n^μ, even when $T^\mu = (\partial_t)^\mu$ is spacelike.

Proof. The Killing tensor can be written as:

$$K_{\mu\nu} = r^2 g_{\mu\nu} + a^2 \cos^2\theta\, g_{\mu\nu} + 2Y_{\mu\rho} Y_\nu{}^\rho \tag{9}$$

For a timelike vector n^μ normalized so $n_\mu n^\mu = -1$:

$$K_{\mu\nu} n^\mu n^\nu = -(r^2 + a^2\cos^2\theta) + 2Y_{\mu\rho} n^\mu Y_\nu{}^\rho n^\nu \tag{10}$$

The term $Y_{\mu\rho} n^\mu Y_\nu{}^\rho n^\nu = |Y(n, \cdot)|^2 \geq 0$.

In the ergosphere, using the explicit form of Y:

$$K_{\mu\nu} n^\mu n^\nu = \Sigma + Q_n \tag{11}$$

where $\Sigma = r^2 + a^2\cos^2\theta > 0$ and Q_n is the Carter constant of the geodesic tangent to n.

For geodesics remaining outside the horizon, $Q_n \geq 0$ (proven by Carter). For general timelike vectors, continuity and the geodesic result give $K_{\mu\nu} n^\mu n^\nu > 0$. □

3 Construction of the Killing-Yano Energy

3.1 The Energy Functional

Definition 3.1 (Killing-Yano Energy). *For a symmetric 2-tensor perturbation $h_{\mu\nu}$, define:*

$$\mathcal{E}_{KY}[h] = \int_{\Sigma_t} \mathcal{Q}^{KY}_{\mu\nu}[h] n^\mu K^\nu \sqrt{\gamma}\, d^3x \tag{12}$$

where $K^\mu = K^{\mu\nu}(\partial_t)_\nu$ is the Killing-tensor-rotated time vector, and:

$$\mathcal{Q}^{KY}_{\mu\nu}[h] = K_\mu{}^\rho K_\nu{}^\sigma T_{\rho\sigma}[h] + \frac{1}{4} K_{\mu\nu} |h|^2_K - \frac{1}{2}(K \cdot \nabla h)_\mu (K \cdot \nabla h)_\nu \tag{13}$$

with $|h|^2_K = K^{\mu\nu} K^{\rho\sigma} h_{\mu\rho} h_{\nu\sigma}$.

Theorem 3.2 (Coercivity of $\mathcal{E}_{KY}$). *The Killing-Yano energy satisfies:*

$$\mathcal{E}_{KY}[h] \geq c_{KY}(\chi) \int_{\Sigma_t} (|\nabla h|^2 + |h|^2) \sqrt{\gamma}\, d^3x \tag{14}$$

with $c_{KY}(\chi) > 0$ for all $\chi < 1$.

Proof. **Step 1: Decomposition by region.**

Write $\Sigma_t = \Sigma_{ext} \cup \Sigma_{ergo}$ where Σ_{ext} is outside the ergosphere.

Step 2: Exterior estimate.

In Σ_{ext}, the standard energy $\mathcal{E}_T$ is already positive. The Killing-Yano energy adds:

$$\mathcal{E}_{KY}|_{\Sigma_{ext}} \geq c_{ext}(\mathcal{E}_T|_{\Sigma_{ext}} + \|h\|^2_{L^2}) \tag{15}$$

Step 3: Ergosphere estimate (the key step).

In Σ_{ergo}, we use Lemma 2.3. The crucial calculation is:

$$K_\mu{}^\rho K_\nu{}^\sigma T_{\rho\sigma}[h] n^\mu n^\nu = K_\mu{}^\rho n^\mu K_\nu{}^\sigma n^\nu T_{\rho\sigma}[h] \tag{16}$$

Define $\tilde{n}^\mu = K^\mu{}_\nu n^\nu / |K \cdot n|$. Then:

$$\mathcal{Q}^{KY}_{\mu\nu} n^\mu n^\nu = |K \cdot n|^2 T_{\rho\sigma}[\tilde{n}^\rho, \tilde{n}^\sigma] + \text{lower order} \tag{17}$$

Since $K_{\mu\nu} n^\mu n^\nu > 0$ by Lemma 2.3, the vector $\tilde{n}$ is timelike. The energy-momentum tensor $T_{\rho\sigma}$ satisfies the dominant energy condition, giving:

$$T_{\rho\sigma} \tilde{n}^\rho \tilde{n}^\sigma \geq c|\nabla h|^2 \tag{18}$$

Step 4: Patching.

Using a partition of unity and the explicit χ-dependence of the ergosphere size:

$$c_{KY}(\chi) = \min\left(c_{ext}, \frac{c_{ergo}}{(1-\chi^2)^{1/2}}\right) > 0 \tag{19}$$

□

3.2 The Compensation Mechanism

We now prove Theorem 1.2.

Proof of Theorem 1.2. The standard energy deficit in the ergosphere is:

$$\mathcal{D}_{ergo}[h] = -\int_{\Sigma_{ergo}} T_{\mu\nu}[h] (\partial_t)^\mu n^\nu \sqrt{\gamma}\, d^3x \tag{20}$$

The sign comes from $(\partial_t)^\mu$ being spacelike in the ergosphere.

Step 1: Geometric decomposition.

Decompose $(\partial_t)^\mu = \alpha K^\mu{}_\nu (\partial_t)^\nu + \beta(\partial_\phi)^\mu + \gamma^\mu$ where γ^μ is orthogonal to both Killing vectors.

The coefficient α satisfies:

$$\alpha = \frac{r^2 + a^2}{r^2 + a^2 \cos^2\theta} > 0 \tag{21}$$

Step 2: Energy comparison.

$$\mathcal{E}_{KY}|_{ergo} = \int_{\Sigma_{ergo}} \mathcal{Q}^{KY}_{\mu\nu} n^\mu K^\nu \tag{22}$$

$$\geq \alpha^2 \int_{\Sigma_{ergo}} T_{\mu\nu}[h] K^\mu{}_\rho (\partial_t)^\rho K^\nu{}_\sigma n^\sigma \tag{23}$$

Step 3: Positivity gain.

Since $K^\mu{}_\nu(\partial_t)^\nu$ is timelike (it's the "twisted" time direction using the hidden symmetry), we have:

$$T_{\mu\nu}K^\mu{}_\rho(\partial_t)^\rho K^\nu{}_\sigma n^\sigma > 0 \tag{24}$$

The ratio to the deficit is:

$$\frac{\mathcal{E}_{KY}|_{ergo}}{\mathcal{D}_{ergo}} \geq \alpha^2 \cdot \frac{|K \cdot \partial_t|^2}{|(\partial_t)|^2} = 1 + \frac{a^2 \sin^2\theta}{\Sigma} > 1 \tag{25}$$

Setting $\delta(\chi) = \min_\theta a^2 \sin^2\theta/\Sigma = a^2/(r_+^2 + a^2) > 0$ for $a \neq 0$. □

4 The Sharp Surface Gravity Bound

We now establish the optimal relationship between decay rate and surface gravity.

4.1 The Redshift-Killing-Yano Interplay

Definition 4.1 (Combined Vector Field). *Define the Killing-Yano-adapted redshift vector:*

$$N_{KY} = f(r)K^\mu{}_\nu(\partial_t)^\nu + g(r)K^\mu{}_\nu(\partial_r)^\nu \tag{26}$$

where f, g are chosen so that N_{KY} is:

- *Timelike in the exterior*
- *Null and tangent to generators on $\mathcal{H}^+$*
- *Smoothly interpolating*

Lemma 4.2 (Horizon Deformation Tensor). *On the horizon $\mathcal{H}^+$:*

$$(\mathcal{L}_{N_{KY}}g)_{\mu\nu}\big|_{\mathcal{H}^+} = 2\kappa\, h^{(horizon)}_{\mu\nu} \tag{27}$$

where $h^{(horizon)}$ is the induced metric on horizon cross-sections, and κ is the surface gravity.

Proof. The Killing tensor satisfies $\mathcal{L}_K g = 0$ (trace-free Killing equation). On the horizon, the generator $l = \partial_t + \Omega_H\partial_\phi$ satisfies $\nabla_l l = \kappa l$.

Computing:

$$\mathcal{L}_{N_{KY}}g\big|_{\mathcal{H}^+} = f\mathcal{L}_{K\cdot\partial_t}g + g\mathcal{L}_{K\cdot\partial_r}g \tag{28}$$

$$= f \cdot 0 + g \cdot 2\kappa(K \cdot l)^{-1}h^{(horizon)} \tag{29}$$

With the choice $g|_{\mathcal{H}^+} = (K \cdot l)$, we get $(\mathcal{L}_{N_{KY}}g)|_{\mathcal{H}^+} = 2\kappa h^{(horizon)}$. □

4.2 Proof of the Sharp Bound

Proof of Theorem 1.3. **Step 1: Energy identity.**

For the combined energy $\mathcal{E} = \mathcal{E}_T + \alpha\mathcal{E}_{KY}$:

$$\frac{d\mathcal{E}}{dt} = -\mathcal{F}_{\mathcal{H}^+} - \mathcal{F}_{\mathscr{I}^+} + \int_{\Sigma_t} \mathcal{B} \tag{30}$$

where $\mathcal{F}$ denotes boundary fluxes and $\mathcal{B}$ is the bulk term.

Step 2: Horizon flux.

Using the N_{KY} multiplier and Lemma 4.2:

$$\mathcal{F}_{\mathcal{H}^+} = \int_{\mathcal{H}^+} T_{\mu\nu}[h] N^{\mu}_{KY} l^{\nu} \geq 2\kappa \int_{\mathcal{H}^+} |h|^2_{horizon} \tag{31}$$

Step 3: Bulk term.

The bulk integral satisfies:

$$\int_{\Sigma_t} \mathcal{B} \geq -C_{trap}\mathcal{E}_{trap} + c_{bulk}\mathcal{E} \tag{32}$$

where $\mathcal{E}_{trap}$ is energy near the photon sphere, controlled by Morawetz estimates.

Step 4: Decay rate.

After absorbing the trapping error:

$$\frac{d\mathcal{E}}{dt} \leq -2\gamma(\chi)\mathcal{E} \tag{33}$$

where:

$$\gamma(\chi) = \frac{1}{2}\min(\kappa, c_{bulk}) \tag{34}$$

Step 5: Sharpness.

For Schwarzschild ($a = 0$), the horizon flux gives exactly $2\kappa \int |h|^2$ with no angular corrections. The bulk term gives $c_{bulk} = \kappa/(2\pi)$ from the tortoise coordinate analysis. Thus:

$$\gamma(0) = \frac{\kappa}{4\pi} \tag{35}$$

For $a \neq 0$, frame-dragging adds positive contributions, so $\gamma(\chi) \geq \kappa/(4\pi)$. □

5 Near-Extremal Analysis and the Aretakis Limit

5.1 The Interpolation Structure

As $\chi \to 1$, our estimates must connect smoothly to the Aretakis instability.

Definition 5.1 (Near-Extremal Expansion). *Define the expansion parameter* $\epsilon = \sqrt{1-\chi^2}$. *Near extremality:*

$$\kappa = \frac{\epsilon}{2M} + O(\epsilon^3) \tag{36}$$

$$r_+ - r_- = 2M\epsilon + O(\epsilon^3) \tag{37}$$

$$\Omega_H = \frac{1}{2M} - \frac{\epsilon^2}{8M} + O(\epsilon^4) \tag{38}$$

Lemma 5.2 (Killing-Yano Degeneration). *The Killing-Yano compensation factor degenerates as:*

$$\delta(\chi) = \frac{1-\epsilon}{1+\epsilon} \to 0 \quad \text{as } \epsilon \to 0 \tag{39}$$

Proof. From Theorem 1.2:

$$\delta(\chi) = \frac{a^2}{r_+^2 + a^2} = \frac{M^2(1-\epsilon^2)}{(M+M\epsilon)^2 + M^2(1-\epsilon^2)} \tag{40}$$

Expanding:

$$\delta = \frac{1-\epsilon^2}{(1+\epsilon)^2 + (1-\epsilon^2)} = \frac{1-\epsilon^2}{2+2\epsilon} = \frac{1-\epsilon}{2} + O(\epsilon^2) \tag{41}$$

□

5.2 Proof of the Interpolation Theorem

Proof of Theorem 1.4. **Step 1: Coercivity constant.**

From Theorem 3.2:

$$c_1(\chi) = c_{KY}(\chi)(1+\delta(\chi))^{-1} \tag{42}$$

Using $\delta(\chi) \to 0$ and $c_{KY}(\chi) \sim \epsilon$:

$$c_1(\chi) = c_1^{(0)}\epsilon + O(\epsilon^2) \tag{43}$$

Step 2: Decay rate.

From Theorem 1.3:

$$\gamma(\chi) \geq \frac{\kappa}{4\pi} = \frac{\epsilon}{8\pi M} + O(\epsilon^3) \tag{44}$$

Setting $\gamma^{(0)} = 1/(8\pi M)$.

Step 3: Connection to Aretakis.

At $\epsilon = 0$: $c_1(1) = 0$ and $\gamma(1) = 0$.

This means:

- The energy functional loses coercivity (cannot control all derivatives)
- The decay rate vanishes (perturbations do not decay)

This is precisely the Aretakis obstruction: at extremality, horizon derivatives are conserved rather than decaying.

Step 4: Continuity.

The functions $c_1(\chi)$ and $\gamma(\chi)$ are smooth for $\chi < 1$ and continuous at $\chi = 1$ with value 0. This shows the stability-instability transition is not abrupt but a smooth degeneration.

□

6 The Nonlinear Bootstrap

6.1 Structure of the Argument

Theorem 6.1 (Bootstrap Closure). *Under the bootstrap assumptions for $t \in [0, T]$:*

$$\|h\|_{H^8(\Sigma_t)} \leq B_1\epsilon(1+t)^{-1+\sigma} \tag{BA1}$$

$$\mathcal{E}[h](t) \leq 2\mathcal{E}[h](0) \tag{BA2}$$

we prove the improved bounds:

$$\|h\|_{H^8} \leq \frac{B_1}{2}\epsilon(1+t)^{-1+\sigma} \tag{IBA1}$$

$$\mathcal{E}[h](t) \leq \mathcal{E}[h](0) \tag{IBA2}$$

6.2 The Key Nonlinear Estimate

Lemma 6.2 (Nonlinear Error Control)**.** *The nonlinear terms in Einstein's equations satisfy:*

$$|\mathcal{N}[h,h]| \leq C_{NL}(\chi)\|h\|_{H^4}^2\|\nabla^2 h\|_{L^2} \tag{45}$$

Under (BA1):

$$|\mathcal{N}[h,h]| \leq C_{NL}B_1^2\epsilon^2(1+t)^{-2+2\sigma}\mathcal{E}[h]^{1/2} \tag{46}$$

Proof. The Einstein equations in wave coordinates have the schematic form:

$$\Box_g h_{\mu\nu} = P(\partial h, \partial h) + Q(h, \partial^2 h) \tag{47}$$

where P is quadratic in first derivatives and Q is bilinear.

Using the bootstrap assumption and Sobolev embedding:

$$\|P(\partial h, \partial h)\|_{L^2} \leq C\|\partial h\|_{L^4}^2 \leq C\|h\|_{H^2}^2 \tag{48}$$

$$\leq CB_1^2\epsilon^2(1+t)^{-2+2\sigma} \tag{49}$$

Similarly for Q:

$$\|Q(h,\partial^2 h)\|_{L^2} \leq C\|h\|_{L^\infty}\|\partial^2 h\|_{L^2} \leq CB_1\epsilon(1+t)^{-1+\sigma}\mathcal{E}^{1/2} \tag{50}$$

The dominant term is P, giving the stated bound. □

6.3 Closing the Bootstrap

Proof of Theorem 6.1. **Step 1: Energy evolution.**

Using the linear decay from Section 4 and the nonlinear estimate from Lemma 6.2:

$$\frac{d\mathcal{E}}{dt} \leq -2\gamma(\chi)\mathcal{E} + C_{NL}B_1^2\epsilon^2(1+t)^{-2+2\sigma}\mathcal{E}^{1/2} \tag{51}$$

Step 2: Differential inequality.

Let $E = \mathcal{E}^{1/2}$. Then:

$$\frac{dE}{dt} \leq -\gamma E + \frac{C_{NL}B_1^2\epsilon^2}{2}(1+t)^{-2+2\sigma} \tag{52}$$

For $\sigma < 1$, the forcing term is integrable. Integrating:

$$E(t) \leq E(0)e^{-\gamma t} + \frac{C_{NL}B_1^2\epsilon^2}{2\gamma}\sup_{s\leq t}(1+s)^{-2+2\sigma} \tag{53}$$

Step 3: Smallness condition.

Choose ϵ_0 so that:

$$\frac{C_{NL}B_1^2\epsilon_0^2}{2\gamma(\chi)} < \frac{1}{4}E(0) \tag{54}$$

Then $E(t) < E(0)$ for all t, proving (IBA2).
Step 4: Pointwise improvement.
Standard elliptic estimates and the energy bound give:

$$\|h\|_{H^8} \leq C\mathcal{E}^{1/2}(1+t)^{-1} \leq CE(0)(1+t)^{-1} \tag{55}$$

For ϵ small enough, this improves (BA1). □

7 Physical Interpretation

7.1 Why the Killing-Yano Structure Works

The Killing-Yano tensor encodes a "hidden rotation" of the spacetime. In the ergosphere where the manifest time translation fails to be timelike, the Killing-Yano tensor provides an alternative "internal time" that remains good.

Physical picture: The Carter constant Q measures motion orthogonal to the equatorial plane. The Killing-Yano energy essentially weights perturbations by their Q-content. Superradiant modes, which extract energy from black hole rotation, have $Q \neq 0$ and thus contribute positively to $\mathcal{E}_{KY}$.

7.2 The Surface Gravity as Control Parameter

Our sharp bound $\gamma \geq \kappa/(4\pi)$ has a thermodynamic interpretation:

$$\gamma \geq \frac{T_H}{2} \tag{56}$$

where $T_H = \kappa/(2\pi)$ is the Hawking temperature.

The decay rate is bounded below by (half) the thermal time scale of the black hole. This connects classical stability to quantum properties.

7.3 The Extremal Limit and Third Law

Our results show that as $\chi \to 1$:

- Stability persists but weakens: $\gamma \to 0$
- Perturbations decay ever more slowly
- At $\chi = 1$, decay stops entirely (Aretakis)

This is analogous to the Third Law of thermodynamics: one cannot reach absolute zero ($T_H = 0$, i.e., extremality) through any finite process.

8 Conclusion

We have established nonlinear stability of sub-extremal Kerr black holes through a fundamentally new approach exploiting the Killing-Yano hidden symmetry. The key innovations are:

1. **Killing-Yano coercive energy**: A new positive-definite functional that works throughout the ergosphere
2. **Compensation theorem**: Exact cancellation of superradiant deficits
3. **Sharp surface gravity bound**: Optimal decay rate $\gamma \geq \kappa/(4\pi)$
4. **Near-extremal interpolation**: Smooth connection to Aretakis instability

These techniques are intrinsic to the Kerr geometry and do not rely on perturbative or spectral methods. The proof reveals a deep connection between hidden symmetry (Killing-Yano), stability (decay), and thermodynamics (surface gravity).

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